Cisco MAR-PE-13 Node Optical Transport Failure SOP 5537

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_5537

Description

Optical transport failures in Cisco PE devices such as MAR-PE-13 at the Marathahalli Core site can be attributed to various factors including SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. These failures are more pronounced due to the inherent characteristics of the Cisco IOS XR software architecture and the physical properties of the optical transport medium. The Cisco PE device, in this case, uses the Cisco IOS XR operating system, which provides a robust framework for managing and monitoring the optical transport network. However, the device's reliance on SFP/XFP modules for optical signal transmission and reception makes it susceptible to degradation and failure. Furthermore, the fiber optic cables connecting the device to other nodes in the network can be prone to cuts and damage, leading to signal loss and network downtime. The alarm code ALARM_5537 indicates a minor severity issue, but prompt attention is required to minimize the impact on network performance.

Pre-requisite Checks

- Confirm that the MAR-PE-13 node is reachable and accessible.

- Verify that the redundant link status is stable and functioning properly.

- Check the environmental conditions at the Marathahalli Core site, including temperature, humidity, and power supply.

- Ensure that the necessary spare parts and tools are available on site.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to optical signal degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces GigabitEthernet0/0/0/0, Expected Output: GigabitEthernet0/0/0/0 is up, line protocol is up, Description: Verify the status of the GigabitEthernet0/0/0/0 interface and ensure it is up and running.

Command: show controllers GigabitEthernet0/0/0/0, Expected Output: GigabitEthernet0/0/0/0 is up, line protocol is up, Description: Display detailed information about the GigabitEthernet0/0/0/0 controller and verify its status.
```

```
Command: show log | include ALARM_5537, Expected Output: ALARM_5537: Minor Optical Transport Failure, Description: Search the system log for the ALARM_5537 message and verify the severity of the issue.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: no service hardware-bundle, Description: Disable the hardware bundle service to prevent bundle-related issues before replacing the faulty SFP/XFP module.

Step 2: Step: Verify the new SFP/XFP module, Command: show controllers GigabitEthernet0/0/0/0, Description: Verify the status of the new SFP/XFP module and ensure it is recognized by the device.

Step 3: Step: Re-enable the hardware bundle service, Command: service hardware-bundle, Description: Re-enable the hardware bundle service to allow bundle-related features to function properly.

Post-Check Verification

```
Command: show interfaces GigabitEthernet0/0/0/0, Expected Output: GigabitEthernet0/0/0/0 is up, line protocol is up, Description: Verify the status of the GigabitEthernet0/0/0/0 interface and ensure it is up and running.

Command: show log | include ALARM_5537, Expected Output: ALARM_5537: Minor Optical Transport Failure, Description: Search the system log for the ALARM_5537 message and verify the severity of the issue.
```

Roll-back Procedures

- Step: Revert to the previous SFP/XFP module, Command: no controllers GigabitEthernet0/0/0/0, Description: Revert to the previous SFP/XFP module by disabling the GigabitEthernet0/0/0/0 controller.

- Step: Re-enable the hardware bundle service, Command: service hardware-bundle, Description: Re-enable the hardware bundle service to allow bundle-related features to function properly.